

ASQ Passes to Symmetric Injective Tensor Powers

Candidate full solution to Question 4.3 of arXiv:1602.07090

Source and Open Question

The source paper is Johann Langemets, Vegard Lima, and Abraham Rueda Zoca, *Almost square and octahedral norms in tensor products of Banach spaces*, arXiv:1602.07090 [1]. In Section 4, Question 4.3, the authors ask:

Let X be a Banach space. Is $\widehat{\otimes}_{\varepsilon,s,N} X$ ASQ for each $N \in \mathbb{N}$ whenever X is ASQ?

The question appears on page 14 of the source PDF:

Corollary 2.7 says that the injective tensor product is ASQ provided one of the factors is ASQ. For the symmetric case we ask:

Question 4.3. *Let X be a Banach space. Is $\widehat{\otimes}_{\varepsilon,s,N} X$ ASQ for each $N \in \mathbb{N}$ whenever X is ASQ?*

Concerning octahedrality in projective tensor products we have only

Definitions

For a Banach space X , the N -fold injective symmetric tensor product $\widehat{\otimes}_{\varepsilon,s,N} X$ is the completion of the linear span of tensors $x^N = x \otimes \cdots \otimes x$ under the norm

$$\left\| \sum_j \lambda_j x_j^N \right\| = \sup_{x^* \in B_{X^*}} \left| \sum_j \lambda_j x^*(x_j)^N \right|.$$

We use the standard equivalent finite-set form of ASQ: a Banach space Z is ASQ if and only if, for every $z_1, \dots, z_m \in S_Z$ and every $\varepsilon > 0$, there is $w \in S_Z$ such that

$$\|z_i \pm w\| \leq 1 + \varepsilon \quad (1 \leq i \leq m).$$

We also use the finite-dimensional ASQ extension theorem of Abrahamsen, Langemets, and Lima [2, Theorem 2.4]: if X is ASQ, then for every finite-dimensional subspace $E \subset X$ and every $\delta > 0$, there is $y \in S_X$ such that

$$\|e + \lambda y\| \leq (1 + \delta) \max\{\|e\|, |\lambda|\} \quad (e \in E, \lambda \in \mathbb{R}).$$

Proof Intuition

The non-symmetric injective tensor proof in the source paper works because an ASQ space contains, near any finite-dimensional piece, a new direction that behaves almost like an ℓ_∞ -summand. The same idea works for the symmetric injective norm once one dualizes the estimate.

Indeed, if $E + \mathbb{R}y$ is almost $E \oplus_\infty \mathbb{R}$, then every $x^* \in B_{X^*}$ satisfies

$$\|x^*|_E\| + |x^*(y)| \leq 1 + \delta.$$

For a finite symmetric tensor supported in E , its value at x^* is bounded by a constant times $\|x^*|_E\|^N$, while the perturbing tensor y^N contributes $|x^*(y)|^N$. The scalar inequality

$$r^N + s^N \leq (r + s)^N$$

then gives exactly the required ASQ estimate.

Lemma 1 (dual estimate). *Let $E \subset X$ be finite-dimensional and let $y \in S_X$ satisfy*

$$\|e + \lambda y\| \leq (1 + \delta) \max\{\|e\|, |\lambda|\} \quad (e \in E, \lambda \in \mathbb{R}).$$

Then every $x^ \in B_{X^*}$ satisfies*

$$\|x^*|_E\| + |x^*(y)| \leq 1 + \delta.$$

Proof. Define $T : E \oplus_\infty \mathbb{R} \rightarrow X$ by $T(e, \lambda) = e + \lambda y$. The hypothesis says that $\|T\| \leq 1 + \delta$. Hence $x^* \circ T$ has norm at most $1 + \delta$ on $E \oplus_\infty \mathbb{R}$. Since $(E \oplus_\infty \mathbb{R})^* = E^* \oplus_1 \mathbb{R}$, this norm is precisely $\|x^*|_E\| + |x^*(y)|$. $\square$

Theorem 1. *Let X be an ASQ Banach space and let $N \in \mathbb{N}$. Then $\widehat{\otimes}_{\varepsilon, s, N} X$ is ASQ.*

Proof. Put $Z = \widehat{\otimes}_{\varepsilon, s, N} X$. Fix $u_1, \dots, u_m \in S_Z$ and $\varepsilon > 0$. Choose $\eta > 0$ and $\delta > 0$ so small that

$$\eta + (1 + \eta)(1 + \delta)^N \leq 1 + \varepsilon.$$

For each i , choose a finite symmetric tensor

$$p_i = \sum_{j=1}^{k_i} \lambda_{ij} x_{ij}^N$$

with $\|u_i - p_i\| < \eta$. Let

$$E = \text{span}\{x_{ij} : 1 \leq i \leq m, 1 \leq j \leq k_i\}.$$

By the ASQ finite-dimensional extension theorem, choose $y \in S_X$ such that

$$\|e + \lambda y\| \leq (1 + \delta) \max\{\|e\|, |\lambda|\} \quad (e \in E, \lambda \in \mathbb{R}).$$

Set $v = y^N \in Z$. Since $\|y\| = 1$, Hahn–Banach gives $\|v\| = 1$.

Fix i and $x^* \in B_{X^*}$. Write

$$r = \|x^*|_E\|, \quad s = |x^*(y)|.$$

By Lemma 1, $r + s \leq 1 + \delta$. The tensor p_i is supported in E , so by Hahn–Banach applied to E ,

$$\left| \sum_{j=1}^{k_i} \lambda_{ij} x^*(x_{ij})^N \right| \leq \|p_i\| r^N.$$

Also $\|p_i\| \leq \|u_i\| + \eta = 1 + \eta$. Therefore, for either choice of sign,

$$\begin{aligned} |p_i(x^*) \pm x^*(y)^N| &\leq \|p_i\| r^N + s^N \\ &\leq (1 + \eta)(r^N + s^N) \\ &\leq (1 + \eta)(r + s)^N \\ &\leq (1 + \eta)(1 + \delta)^N. \end{aligned}$$

Taking the supremum over $x^* \in B_{X^*}$ yields

$$\|p_i \pm v\| \leq (1 + \eta)(1 + \delta)^N.$$

Consequently,

$$\|u_i \pm v\| \leq \|u_i - p_i\| + \|p_i \pm v\| < \eta + (1 + \eta)(1 + \delta)^N \leq 1 + \varepsilon.$$

This holds for every $i = 1, \dots, m$, so Z is ASQ. $\square$

Verification Notes

The proof uses no computation. The included script `code/scalar_bound_verifier.py` only audits the scalar inequality $r^N + s^N \leq (r + s)^N$ on grids and random samples; the analytic reason is the binomial theorem for $r, s \geq 0$.

The finite-tensor approximation step is legitimate because $\widehat{\otimes}_{\varepsilon, s, N} X$ is defined as the completion of $\otimes^{s, N} X$. The estimate for a finite tensor supported in E uses the equality of the injective symmetric norm with the supremum over B_{E^*} , which follows from Hahn–Banach extension from E to X .

Novelty Check and Limitations

Local duplicate checks in the registry, solution, attempt, and proof-gap indexes, together with the target-pool scan, found no previous packet for arXiv:1602.07090 or for Question 4.3. External searches on 2026-06-19 for exact phrases including “injective symmetric tensor product”, “almost square”, “ASQ”, “ $\widehat{\otimes}_{\varepsilon, s, N}$ ”, and “1602.07090” surfaced the source paper but no later explicit answer. This packet should therefore be treated as a plausible new affirmative solution, subject to human literature review.

Human Review Recommendation

Recommended review focus: verify Lemma 1, the coefficient-safe finite tensor estimate, and the completion argument. If these pass, the packet gives a full affirmative answer to Question 4.3 of the source paper.

References

- [1] J. Langemets, V. Lima, and A. Rueda Zoca, *Almost square and octahedral norms in tensor products of Banach spaces*, arXiv:1602.07090, 2016.
- [2] T. A. Abrahamsen, J. Langemets, and V. Lima, *Almost square Banach spaces*, J. Math. Anal. Appl. 434 (2016), 1549–1565.